# Three-$\rho$ terms $\propto \log(\vee/\Lambda)$: they cancel exactly

by kernel class $(\alpha, \alpha', \beta, \beta')$ and number of Wilson lines

## Result

- **All three-$\rho$ terms cancel exactly.** The list has 152 terms. After symmetrisation of the charges they form 40 distinct colour networks, and the coefficients of every network add up to zero.
- **The cancellation is term against term.** In every network the cancelling terms have literally the same colour structure: the same charges, rotated by the same chain of Wilson lines, up to the order of the indices of $f^{abc}$ (a sign). No identity is needed: neither $U^TU = 1$ nor $f^{abc}U^{aa'}U^{bb'}U^{cc'} = f^{a'b'c'}$.
- **Each class and each number of Wilson lines cancels separately.** The four kernel classes $\alpha, \alpha', \beta, \beta'$ do not mix. Within a class every network contains terms with the same number of Wilson lines, so the two-, three- and four-Wilson-line terms each cancel among themselves (Secs. 1–4).
- **Numerical check.** With random SU(3) Wilson lines and charges, the residual of every kernel monomial is below $3 \times 10^{-15}$, while single terms are of order one (Sec. 5).
- **Same as the corrected list.** Group by group, the list equals the corrected list of `ThreeRho_final.pdf` (T1–T3, F3, F4 applied).
- **Convention $U^{bc} = U^{cb}$.** The list is written with your convention. Read literally with a general (non-symmetric) adjoint $U$, the sums do not vanish. They do vanish for a general $U$ if every chain is read as $U(p_1)\cdots U(p_n)\rho$, with each $U$ oriented from $f$ towards the charge. So the cancellation does not rely on $U^2 = 1$; only the index order of the written placements assumes $U = U^T$.

## Conventions

- All terms carry the common factor $\mathcal{N} \equiv \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}$. The four-$\rho$ block $d^3N_a$ has $-\mathcal{N}$ (its prefactor is $-\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{\log}{2\pi^2}$), $d^3N_b$ has $+\mathcal{N}$.
- The coefficient $c$ of a term (in units of $\mathcal{N}$) is its sign $\times 2$ (for "2") $\times \frac{1}{2}$ (for "$\frac{1}{2}$") $\times 2$ (for an anticommutator). Only the fully symmetrised part of the charges is compared, so $\frac{1}{2}\rho\{\rho,\rho\}$ counts as $\rho\rho\rho$ and $\{\rho,\rho\}\rho$ as $2\rho\rho\rho$.
- Within a class the kernel is common to all terms, and the charges sit at $x$, $x'$, $y$ with fixed roles. The terms of a class cancel if and only if their colour factors, as functions of $x, x', y, w, w', z$, add up to zero.
- A colour network is written $f^{abc}L_1^aL_2^bL_3^c$. Each leg $L$ is a charge rotated by a chain of Wilson lines, e.g. $\big[U(w)U(w')\rho(x')\big]^c = U^{cd}(w)U^{de}(w')\rho^e(x')$ (with $U^{bc} = U^{cb}$). A term written in another index order is mapped to its network by reordering $f^{abc}$, which gives the sign $s = \pm 1$; its contribution to the network is $c\,s$.
- Labels: $4\rho_a$, $4\rho_b$ = from four $\rho$'s ($d^3N_a$, $d^3N_b$); I.II.1 = Group I Row II part 1, …; III.VI = Group III Row VI. The number in brackets is the position of the term in its block, as sent.

- In each subsection, every network has its own colour. Its terms are listed together, and the last line of each network is the sum of their contributions.

## Summary

| class | Wilson lines | terms | networks | sum of every network |
|---|---|---|---|---|
| $\alpha$ | 2 | 4 | 2 | 0 (all) |
| $\alpha$ | 3 | 18 | 4 | 0 (all) |
| $\alpha$ | 4 | 6 | 2 | 0 (all) |
| $\alpha'$ | 2 | 4 | 2 | 0 (all) |
| $\alpha'$ | 3 | 18 | 4 | 0 (all) |
| $\alpha'$ | 4 | 6 | 2 | 0 (all) |
| $\beta$ | 2 | 6 | 2 | 0 (all) |
| $\beta$ | 3 | 36 | 8 | 0 (all) |
| $\beta$ | 4 | 6 | 2 | 0 (all) |
| $\beta'$ | 2 | 6 | 2 | 0 (all) |
| $\beta'$ | 3 | 36 | 8 | 0 (all) |
| $\beta'$ | 4 | 6 | 2 | 0 (all) |
| total | | 152 | 40 | |

# 1 $\alpha$ terms

Kernel: $\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$. 28 terms, 8 networks.

## 1.1 Two Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\rho^a(x)\,\rho^b(y)\,\big[U(w)U(w')\rho(x')\big]^c$ | | | | |
| $\mathsf{N}_1$ | $4\rho_a\,(1)$ | $-1$ | $f^{abc}\rho^d(x')\rho^a(y)\rho^c(x)\,U^{de}(w')U^{eb}(w)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | II.I (1) | $-1$ | $f^{abc}\,\{\rho^e(x'),\rho^b(y)\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$ | $-$ | $+1$ |
| | | | | **sum for $\mathsf{N}_1$:** | $-1+1=0$ |
| $\mathsf{N}_2 =$ | $f^{abc}\,\rho^a(x)\,\rho^b(y)\,\big[U(w)U(x')\rho(x')\big]^c$ | | | | |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_2$ | $4\rho_a\,(2)$ | $+1$ | $f^{abc}\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{ed}(x')$ | $+$ | $+1$ |
| $\mathsf{N}_2$ | III.II (3) | $+1$ | $f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)$ | $-$ | $-1$ |
| | | | | **sum for $\mathsf{N}_2$:** | $1-1=0$ |

## 1.2 Three Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\left[U(w')\rho(x')\right]^a\left[U(x)\rho(x)\right]^b\left[U(y)\rho(y)\right]^c$ | | | | |
| $\mathsf{N}_1$ | $4\rho_b\,(1)$ | $-1$ | $f^{abc}\rho^d(x')\rho^h(y)\rho^e(x)U^{db}(w')U^{ec}(x)U^{ah}(y)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | I.II.1 (3) | $+1$ | $f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)$ | $+$ | $+1$ |
| $\mathsf{N}_1$ | I.II.1 (4) | $-1$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)$ | $-$ | $+1$ |
| $\mathsf{N}_1$ | I.III (4) | $+1$ | $f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)$ | $-$ | $-1$ |
| $\mathsf{N}_1$ | I.IV (4) | $+1$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{ch}(x)$ | $-$ | $-1$ |
| $\mathsf{N}_1$ | III.I (4) | $-1$ | $f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)$ | $-$ | $+1$ |
| | | | | **sum for $\mathsf{N}_1$:** | $-1+1+1-1-1+1=0$ |
| $\mathsf{N}_2 =$ | $f^{abc}\left[U(x)\rho(x)\right]^a\left[U(x')\rho(x')\right]^b\left[U(y)\rho(y)\right]^c$ | | | | |
| $\mathsf{N}_2$ | $4\rho_b\,(4)$ | $+1$ | $f^{abc}\rho^e(x')\rho^h(y)\rho^d(x)U^{dc}(x)U^{be}(x')U^{ah}(y)$ | $-$ | $-1$ |
| $\mathsf{N}_2$ | I.II.1 (1) | $-1$ | $f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)$ | $-$ | $+1$ |
| $\mathsf{N}_2$ | I.II.1 (2) | $+1$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$ | $+$ | $+1$ |
| $\mathsf{N}_2$ | I.III (2) | $-1$ | $f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)$ | $+$ | $-1$ |
| $\mathsf{N}_2$ | I.IV (2) | $-1$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)$ | $+$ | $-1$ |
| $\mathsf{N}_2$ | II.I (4) | $-1$ | $f^{abc}\left\{\rho^d(x'),\rho^e(y)\right\}\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$ | $+$ | $-1$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_2$ | III.I (2) | $+1$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$ | $+$ | $+1$ |
| $\mathsf{N}_2$ | III.II (2) | $+1$ | $f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)$ | $+$ | $+1$ |
| | | | | **sum for $\mathsf{N}_2$:** | $-1+1+1-1-1-1+1+1=0$ |
| $\mathsf{N}_3 =$ | $f^{abc}\,\big[U(z)\rho(y)\big]^a\,\big[U(x)\rho(x)\big]^b\,\big[U(x')\rho(x')\big]^c$ | | | | |
| $\mathsf{N}_3$ | I.III (1) | $+1$ | $f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)$ | $+$ | $+1$ |
| $\mathsf{N}_3$ | III.II (4) | $-1$ | $f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)$ | $+$ | $-1$ |
| | | | | **sum for $\mathsf{N}_3$:** | $1-1=0$ |
| $\mathsf{N}_4 =$ | $f^{abc}\,\big[U(w')\rho(x')\big]^a\,\big[U(z)\rho(y)\big]^b\,\big[U(x)\rho(x)\big]^c$ | | | | |
| $\mathsf{N}_4$ | I.III (3) | $-1$ | $f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)$ | $+$ | $-1$ |
| $\mathsf{N}_4$ | II.I (2) | $+1$ | $f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)$ | $+$ | $+1$ |
| | | | | **sum for $\mathsf{N}_4$:** | $-1+1=0$ |

## 1.3 Four Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\rho^a(x)\,\big[U(w)U(x')\rho(x')\big]^b\,\big[U(z)U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_1$ | I.IV (1) | $+1$ | $f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)$ | $+$ | $+1$ |
| $\mathsf{N}_1$ | II.I (3) | $+1$ | $f^{abc}\,\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ae}(w)$ | $+$ | $+1$ |
| $\mathsf{N}_1$ | III.I (1) | $-1$ | $f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | III.II (1) | $-1$ | $f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ah}(w)$ | $+$ | $-1$ |
| | | | | **sum for $\mathsf{N}_1$:** | $1+1-1-1=0$ |
| $\mathsf{N}_2 =$ | $f^{abc}\,\rho^a(x)\,\big[U(w)U(w')\rho(x')\big]^b\,\big[U(z)U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_2$ | I.IV (3) | $-1$ | $f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)$ | $+$ | $-1$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_2$ | III.I (3) | $+1$ | $f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)$ | $+$ | $+1$ |
| | | | | **sum for $\mathsf{N}_2$:** | $-1+1=0$ |

## 2 $\alpha'$ terms

Kernel: $\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$. 28 terms, 8 networks.

### 2.1 Two Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\rho^a(x')\,\rho^b(y)\,\big[U(w')U(x)\rho(x)\big]^c$ | | | | |
| $\mathsf{N}_1$ | $4\rho_a$ (3) | $-1$ | $f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{ed}(x)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | III.II (11) | $-1$ | $f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{ad}(w')$ | $-$ | $+1$ |
| | | | | **sum for $\mathsf{N}_1$:** | $-1+1=0$ |
| $\mathsf{N}_2 =$ | $f^{abc}\,\rho^a(x')\,\rho^b(y)\,\big[U(w')U(w)\rho(x)\big]^c$ | | | | |
| $\mathsf{N}_2$ | $4\rho_a$ (4) | $+1$ | $f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)U^{bd}(w')U^{ed}(w)$ | $+$ | $+1$ |
| $\mathsf{N}_2$ | II.I (9) | $+1$ | $f^{abc}\,\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{ad}(w')\,U^{de}(w)$ | $-$ | $-1$ |
| | | | | **sum for $\mathsf{N}_2$:** | $1-1=0$ |

### 2.2 Three Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\big[U(w)\rho(x)\big]^a\,\big[U(x')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_1$ | $4\rho_b$ (2) | $+1$ | $f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{eb}(w)U^{ah}(y)$ | $+$ | $+1$ |
| $\mathsf{N}_1$ | I.II.2 (3) | $-1$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | I.II.2 (4) | $+1$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')$ | $-$ | $-1$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1$ | I.III (8) | $-1$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)$ | $-$ | $+1$ |
| $\mathsf{N}_1$ | I.IV (8) | $-1$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(w)\,U^{ch}(x')\,U^{be}(y)$ | $-$ | $+1$ |
| $\mathsf{N}_1$ | III.I (12) | $+1$ | $f^{abc}\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')$ | $-$ | $-1$ |
| | | | | **sum for $\mathsf{N}_1$:** $1-1-1+1+1-1=0$ | |
| $\mathsf{N}_2 = f^{abc}\left[U(x)\rho(x)\right]^a\left[U(x')\rho(x')\right]^b\left[U(y)\rho(y)\right]^c$ | | | | | |
| $\mathsf{N}_2$ | $4\rho_b$ (3) | $-1$ | $f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{be}(x)U^{ah}(y)$ | $+$ | $-1$ |
| $\mathsf{N}_2$ | I.II.2 (1) | $+1$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)$ | $+$ | $+1$ |
| $\mathsf{N}_2$ | I.II.2 (2) | $-1$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')$ | $-$ | $+1$ |
| $\mathsf{N}_2$ | I.III (6) | $+1$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)$ | $-$ | $-1$ |
| $\mathsf{N}_2$ | I.IV (6) | $+1$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(x)\,U^{ch}(x')\,U^{be}(y)$ | $-$ | $-1$ |
| $\mathsf{N}_2$ | II.I (12) | $+1$ | $f^{abc}\,\rho^d(x')\{\rho^e(x),\rho^h(y)\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)$ | $-$ | $-1$ |
| $\mathsf{N}_2$ | III.I (10) | $-1$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')$ | $-$ | $+1$ |
| $\mathsf{N}_2$ | III.II (10) | $-1$ | $f^{abc}\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')$ | $-$ | $+1$ |
| | | | | **sum for $\mathsf{N}_2$:** $-1+1+1-1-1-1+1+1=0$ | |
| $\mathsf{N}_3 = f^{abc}\left[U(z)\rho(y)\right]^a\left[U(x)\rho(x)\right]^b\left[U(x')\rho(x')\right]^c$ | | | | | |
| $\mathsf{N}_3$ | I.III (5) | $-1$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)$ | $-$ | $+1$ |
| $\mathsf{N}_3$ | III.II (12) | $+1$ | $f^{abc}\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')$ | $-$ | $-1$ |
| | | | | **sum for $\mathsf{N}_3$:** $1-1=0$ | |
| $\mathsf{N}_4 = f^{abc}\left[U(w)\rho(x)\right]^a\left[U(z)\rho(y)\right]^b\left[U(x')\rho(x')\right]^c$ | | | | | |
| $\mathsf{N}_4$ | I.III (7) | $+1$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)$ | $+$ | $+1$ |
| $\mathsf{N}_4$ | II.I (11) | $-1$ | $f^{abc}\,\rho^e(x')\{\rho^h(x),\rho^d(y)\}\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)$ | $+$ | $-1$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| | | | | **sum for $\mathsf{N}_4$:** | $1-1=0$ |

### 2.3 Four Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\rho^a(x')\,\big[U(w')U(x)\rho(x)\big]^b\,\big[U(z)U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_1$ | I.IV (5) | $-1$ | $f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | II.I (10) | $-1$ | $f^{abc}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | III.I (9) | $+1$ | $f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')$ | $+$ | $+1$ |
| $\mathsf{N}_1$ | III.II (9) | $+1$ | $f^{abc}\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ah}(w')$ | $+$ | $+1$ |
| | | | | **sum for $\mathsf{N}_1$:** | $-1-1+1+1=0$ |
| $\mathsf{N}_2 =$ | $f^{abc}\,\rho^a(x')\,\big[U(w')U(w)\rho(x)\big]^b\,\big[U(z)U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_2$ | I.IV (7) | $+1$ | $f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)$ | $+$ | $+1$ |
| $\mathsf{N}_2$ | III.I (11) | $-1$ | $f^{abc}\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')$ | $+$ | $-1$ |
| | | | | **sum for $\mathsf{N}_2$:** | $1-1=0$ |

## 3 $\beta$ terms

Kernel: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$. 48 terms, 12 networks.

### 3.1 Two Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\rho^a(x)\,\rho^b(y)\,\big[U(w)U(x')\rho(x')\big]^c$ | | | | |
| $\mathsf{N}_1$ | I.II.1 (5) | $-1$ | $f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{ad}(w)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | I.II.1 (6) | $+1$ | $f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)$ | $-$ | $-1$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1$ | III.II (7) | $-2$ | $f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)$ | $-$ | $+2$ |
| | | | | **sum for $\mathsf{N}_1$:** | $-1-1+2=0$ |
| $\mathsf{N}_2 =$ | $f^{abc}\,\rho^a(x)\,\rho^b(y)\,\big[U(w)U(w')\rho(x')\big]^c$ | | | | |
| $\mathsf{N}_2$ | I.II.1 (9) | $+1$ | $f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{ad}(w)$ | $+$ | $+1$ |
| $\mathsf{N}_2$ | I.II.1 (10) | $-1$ | $f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$ | $-$ | $+1$ |
| $\mathsf{N}_2$ | II.I (5) | $+2$ | $f^{abc}\,\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$ | $-$ | $-2$ |
| | | | | **sum for $\mathsf{N}_2$:** | $1+1-2=0$ |

### 3.2 Three Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\big[U(x)\rho(x)\big]^a\,\big[U(x')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_1$ | I.II.1 (7) | $+1$ | $f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)$ | $-$ | $-1$ |
| $\mathsf{N}_1$ | I.II.1 (8) | $-1$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | I.III (10) | $+2$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)$ | $-$ | $-2$ |
| $\mathsf{N}_1$ | I.IV (11) | $+2$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{hc}(x)$ | $+$ | $+2$ |
| $\mathsf{N}_1$ | I.V.1 (3) | $+1$ | $f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)$ | $+$ | $+1$ |
| $\mathsf{N}_1$ | I.V.2 (2) | $-1$ | $f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)$ | $-$ | $+1$ |
| $\mathsf{N}_1$ | II.I (8) | $+2$ | $f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)$ | $+$ | $+2$ |
| $\mathsf{N}_1$ | III.I (6) | $-2$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)$ | $+$ | $-2$ |
| $\mathsf{N}_1$ | III.II (6) | $-2$ | $f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{db}(y)\,U^{ae}(x')\,U^{ch}(x)$ | $+$ | $-2$ |
| $\mathsf{N}_1$ | III.III (4) | $+2$ | $f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)$ | $-$ | $-2$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1$ | III.V (3) | $-2$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')$ | $-$ | $+2$ |
| $\mathsf{N}_1$ | III.VI (3) | $-2$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{bh}(y)\,U^{ad}(x)\,U^{ce}(x')$ | $-$ | $+2$ |
| | | | | **sum for $\mathsf{N}_1$:** $-1-1-2+2+1+1+2-2-2-2+2+2=0$ | |
| $\mathsf{N}_2 =$ | $f^{abc}\,\big[U(w')\rho(x')\big]^a\,\big[U(x)\rho(x)\big]^b\,\big[U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_2$ | I.II.1 (11) | $-1$ | $f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)$ | $+$ | $-1$ |
| $\mathsf{N}_2$ | I.II.1 (12) | $+1$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)$ | $-$ | $-1$ |
| $\mathsf{N}_2$ | I.III (12) | $-2$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)$ | $+$ | $-2$ |
| $\mathsf{N}_2$ | I.IV (12) | $-2$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{hc}(x)$ | $-$ | $+2$ |
| $\mathsf{N}_2$ | I.V.1 (4) | $-1$ | $f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)$ | $-$ | $+1$ |
| $\mathsf{N}_2$ | I.V.2 (4) | $+1$ | $f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)$ | $+$ | $+1$ |
| $\mathsf{N}_2$ | III.I (8) | $+2$ | $f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)$ | $-$ | $-2$ |
| $\mathsf{N}_2$ | III.V (4) | $+2$ | $f^{abc}\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(w')\,U^{bh}(y)$ | $+$ | $+2$ |
| | | | | **sum for $\mathsf{N}_2$:** $-1-1-2+2+1+1-2+2=0$ | |
| $\mathsf{N}_3 =$ | $f^{abc}\,\big[U(w)\rho(x)\big]^a\,\big[U(x')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_3$ | I.III (9) | $-2$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(w)$ | $-$ | $+2$ |
| $\mathsf{N}_3$ | III.III (2) | $-2$ | $f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)$ | $-$ | $+2$ |
| $\mathsf{N}_3$ | III.V (1) | $+2$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')$ | $-$ | $-2$ |
| $\mathsf{N}_3$ | III.VI (1) | $+2$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)$ | $-$ | $-2$ |
| | | | | **sum for $\mathsf{N}_3$:** $2+2-2-2=0$ | |
| $\mathsf{N}_4 =$ | $f^{abc}\,\big[U(w)\rho(x)\big]^a\,\big[U(w')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_4$ | I.III (11) | $+2$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(w)$ | $-$ | $-2$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_4$ | III.V (2) | $-2$ | $f^{abc}\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(w')\,U^{bh}(y)$ | $-$ | $+2$ |
| | | | | **sum for $\mathsf{N}_4$:** $-2+2=0$ | |
| $\mathsf{N}_5 =$ | $f^{abc}\,\big[U(w)\rho(x)\big]^a\,\big[U(z)\rho(y)\big]^b\,\big[U(x')\rho(x')\big]^c$ | | | | |
| $\mathsf{N}_5$ | I.V.1 (1) | $-1$ | $f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)$ | $-$ | $+1$ |
| $\mathsf{N}_5$ | I.V.2 (1) | $+1$ | $f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z)$ | $+$ | $+1$ |
| $\mathsf{N}_5$ | III.VI (2) | $-2$ | $f^{abc}\,\rho^e(y)\,\rho^h(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(x')$ | $+$ | $-2$ |
| | | | | **sum for $\mathsf{N}_5$:** $1+1-2=0$ | |
| $\mathsf{N}_6 =$ | $f^{abc}\,\big[U(w)\rho(x)\big]^a\,\big[U(w')\rho(x')\big]^b\,\big[U(z)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_6$ | I.V.1 (2) | $+1$ | $f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)$ | $+$ | $+1$ |
| $\mathsf{N}_6$ | I.V.2 (3) | $-1$ | $f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z)$ | $-$ | $+1$ |
| $\mathsf{N}_6$ | III.III (1) | $+2$ | $f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')$ | $-$ | $-2$ |
| | | | | **sum for $\mathsf{N}_6$:** $1+1-2=0$ | |
| $\mathsf{N}_7 =$ | $f^{abc}\,\big[U(w')\rho(x')\big]^a\,\big[U(z)\rho(y)\big]^b\,\big[U(x)\rho(x)\big]^c$ | | | | |
| $\mathsf{N}_7$ | II.I (6) | $-2$ | $f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)$ | $+$ | $-2$ |
| $\mathsf{N}_7$ | III.III (3) | $-2$ | $f^{abc}\,\big\{\rho^h(x'),\rho^d(y)\big\}\rho^e(x)\,U^{db}(z)\,U^{ae}(x)\,U^{ch}(w')$ | $-$ | $+2$ |
| | | | | **sum for $\mathsf{N}_7$:** $-2+2=0$ | |
| $\mathsf{N}_8 =$ | $f^{abc}\,\big[U(z)\rho(y)\big]^a\,\big[U(x)\rho(x)\big]^b\,\big[U(x')\rho(x')\big]^c$ | | | | |
| $\mathsf{N}_8$ | III.II (8) | $+2$ | $f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)$ | $+$ | $+2$ |
| $\mathsf{N}_8$ | III.VI (4) | $+2$ | $f^{abc}\,\rho^d(y)\,\rho^h(x')\,\rho^e(x)\,U^{ae}(x)\,U^{ch}(x')\,U^{db}(z)$ | $-$ | $-2$ |
| | | | | **sum for $\mathsf{N}_8$:** $2-2=0$ | |

### 3.3 Four Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\rho^a(x)\,\big[U(w)U(x')\rho(x')\big]^b\,\big[U(z)U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_1$ | I.IV (9) | $-2$ | $f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)$ | $+$ | $-2$ |
| $\mathsf{N}_1$ | II.I (7) | $-2$ | $f^{abc}\,\{\rho^h(x'),\rho^l(y)\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ea}(w)$ | $+$ | $-2$ |
| $\mathsf{N}_1$ | III.I (5) | $+2$ | $f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)$ | $+$ | $+2$ |
| $\mathsf{N}_1$ | III.II (5) | $+2$ | $f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ha}(w)$ | $+$ | $+2$ |
| | | | | **sum for $\mathsf{N}_1$:** | $-2-2+2+2=0$ |
| $\mathsf{N}_2 =$ | $f^{abc}\,\rho^a(x)\,\big[U(w)U(w')\rho(x')\big]^b\,\big[U(z)U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_2$ | I.IV (10) | $+2$ | $f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)$ | $+$ | $+2$ |
| $\mathsf{N}_2$ | III.I (7) | $-2$ | $f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)$ | $+$ | $-2$ |
| | | | | **sum for $\mathsf{N}_2$:** | $2-2=0$ |

# 4 $\beta'$ terms

Kernel: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$. 48 terms, 12 networks.

## 4.1 Two Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\rho^a(x')\,\rho^b(y)\,\big[U(w')U(x)\rho(x)\big]^c$ | | | | |
| $\mathsf{N}_1$ | I.II.2 (5) | $+1$ | $f^{abc}\,\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')$ | $+$ | $+1$ |
| $\mathsf{N}_1$ | I.II.2 (6) | $-1$ | $f^{abc}\,\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')$ | $-$ | $+1$ |
| $\mathsf{N}_1$ | III.II (15) | $+2$ | $f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{ad}(w')$ | $-$ | $-2$ |
| | | | | **sum for $\mathsf{N}_1$:** | $1+1-2=0$ |
| $\mathsf{N}_2 =$ | $f^{abc}\,\rho^a(x')\,\rho^b(y)\,\big[U(w')U(w)\rho(x)\big]^c$ | | | | |
| $\mathsf{N}_2$ | I.II.2 (9) | $-1$ | $f^{abc}\,\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')$ | $+$ | $-1$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_2$ | I.II.2 (10) | $+1$ | $f^{abc}\,\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')$ | $-$ | $-1$ |
| $\mathsf{N}_2$ | II.I (13) | $-2$ | $f^{abc}\,\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{ad}(w')\,U^{de}(w)$ | $-$ | $+2$ |
| | | | | **sum for $\mathsf{N}_2$:** | $-1-1+2=0$ |

## 4.2 Three Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\big[U(x)\rho(x)\big]^a\,\big[U(x')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_1$ | I.II.2 (7) | $-1$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)$ | $+$ | $-1$ |
| $\mathsf{N}_1$ | I.II.2 (8) | $+1$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')$ | $-$ | $-1$ |
| $\mathsf{N}_1$ | I.III (14) | $-2$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')$ | $+$ | $-2$ |
| $\mathsf{N}_1$ | I.IV (16) | $-2$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')$ | $-$ | $+2$ |
| $\mathsf{N}_1$ | I.V.1 (8) | $-1$ | $f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{bh}(y)\,U^{ce}(x')$ | $-$ | $+1$ |
| $\mathsf{N}_1$ | I.V.2 (6) | $+1$ | $f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')$ | $+$ | $+1$ |
| $\mathsf{N}_1$ | II.I (16) | $-2$ | $f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)$ | $-$ | $+2$ |
| $\mathsf{N}_1$ | III.I (14) | $+2$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')$ | $-$ | $-2$ |
| $\mathsf{N}_1$ | III.II (14) | $+2$ | $f^{abc}\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')$ | $-$ | $-2$ |
| $\mathsf{N}_1$ | III.III (8) | $-2$ | $f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)$ | $+$ | $-2$ |
| $\mathsf{N}_1$ | III.V (7) | $+2$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{be}(y)\,U^{ad}(x')\,U^{ch}(x)$ | $+$ | $+2$ |
| $\mathsf{N}_1$ | III.VI (7) | $+2$ | $f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{bh}(y)\,U^{ad}(x')\,U^{ce}(x)$ | $+$ | $+2$ |
| | | | | **sum for $\mathsf{N}_1$:** | $-1-1-2+2+1+1+2-2-2-2+2+2=0$ |

$\mathsf{N}_2 =\; f^{abc}\,\big[U(w)\rho(x)\big]^a\,\big[U(x')\rho(x')\big]^b\,\big[U(y)\rho(y)\big]^c$

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_2$ | I.II.2 (11) | $+1$ | $f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)$ | $+$ | $+1$ |
| $\mathsf{N}_2$ | I.II.2 (12) | $-1$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')$ | $-$ | $+1$ |
| $\mathsf{N}_2$ | I.III (16) | $+2$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')$ | $+$ | $+2$ |
| $\mathsf{N}_2$ | I.IV (15) | $+2$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(w)\,U^{ch}(x')$ | $-$ | $-2$ |
| $\mathsf{N}_2$ | I.V.1 (7) | $+1$ | $f^{abc}\,\{\rho^e(x'),\rho^h(y)\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(y)\,U^{ce}(x')$ | $-$ | $-1$ |
| $\mathsf{N}_2$ | I.V.2 (8) | $-1$ | $f^{abc}\,\{\rho^h(x'),\rho^e(y)\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(x')\,U^{ce}(y)$ | $+$ | $-1$ |
| $\mathsf{N}_2$ | III.I (16) | $-2$ | $f^{abc}\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')$ | $-$ | $+2$ |
| $\mathsf{N}_2$ | III.V (8) | $-2$ | $f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(x')\,U^{ce}(w)\,U^{bh}(y)$ | $+$ | $-2$ |
| | | | | **sum for $\mathsf{N}_2$:** | $1+1+2-2-1-1+2-2=0$ |
| $\mathsf{N}_3 =$ | $f^{abc}\,[U(w')\rho(x')]^a\,[U(x)\rho(x)]^b\,[U(y)\rho(y)]^c$ | | | | |
| $\mathsf{N}_3$ | I.III (13) | $+2$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w')$ | $-$ | $-2$ |
| $\mathsf{N}_3$ | III.III (6) | $+2$ | $f^{abc}\,\rho^d(x')\{\rho^e(x),\rho^h(y)\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)$ | $-$ | $-2$ |
| $\mathsf{N}_3$ | III.V (5) | $-2$ | $f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)$ | $-$ | $+2$ |
| $\mathsf{N}_3$ | III.VI (5) | $-2$ | $f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{bh}(y)\,U^{ce}(x)$ | $-$ | $+2$ |
| | | | | **sum for $\mathsf{N}_3$:** | $-2-2+2+2=0$ |
| $\mathsf{N}_4 =$ | $f^{abc}\,[U(w)\rho(x)]^a\,[U(w')\rho(x')]^b\,[U(y)\rho(y)]^c$ | | | | |
| $\mathsf{N}_4$ | I.III (15) | $-2$ | $f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w')$ | $+$ | $-2$ |
| $\mathsf{N}_4$ | III.V (6) | $+2$ | $f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(w')\,U^{ce}(w)\,U^{bh}(y)$ | $+$ | $+2$ |
| | | | | **sum for $\mathsf{N}_4$:** | $-2+2=0$ |
| $\mathsf{N}_5 =$ | $f^{abc}\,[U(w')\rho(x')]^a\,[U(z)\rho(y)]^b\,[U(x)\rho(x)]^c$ | | | | |
| $\mathsf{N}_5$ | I.V.1 (5) | $+1$ | $f^{abc}\,\{\rho^h(x'),\rho^e(y)\}\rho^d(x)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')$ | $-$ | $-1$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_5$ | I.V.2 (5) | $-1$ | $f^{abc}\{\rho^e(x'),\rho^h(y)\}\rho^d(x)\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z)$ | $+$ | $-1$ |
| $\mathsf{N}_5$ | III.VI (6) | $+2$ | $f^{abc}\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x)$ | $+$ | $+2$ |
| | | | | **sum for $\mathsf{N}_5$:** | $-1-1+2=0$ |
| $\mathsf{N}_6 =$ | $f^{abc}\,[U(w)\rho(x)]^a\,[U(w')\rho(x')]^b\,[U(z)\rho(y)]^c$ | | | | |
| $\mathsf{N}_6$ | I.V.1 (6) | $-1$ | $f^{abc}\{\rho^h(x'),\rho^e(y)\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')$ | $-$ | $+1$ |
| $\mathsf{N}_6$ | I.V.2 (7) | $+1$ | $f^{abc}\{\rho^e(x'),\rho^h(y)\}\rho^d(x)\,U^{ad}(w)\,U^{ch}(z)\,U^{be}(w')$ | $+$ | $+1$ |
| $\mathsf{N}_6$ | III.III (5) | $-2$ | $f^{abc}\,\rho^d(x')\{\rho^h(x),\rho^e(y)\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)$ | $+$ | $-2$ |
| | | | | **sum for $\mathsf{N}_6$:** | $1+1-2=0$ |
| $\mathsf{N}_7 =$ | $f^{abc}\,[U(w)\rho(x)]^a\,[U(z)\rho(y)]^b\,[U(x')\rho(x')]^c$ | | | | |
| $\mathsf{N}_7$ | II.I (15) | $+2$ | $f^{abc}\,\rho^e(x')\{\rho^h(x),\rho^d(y)\}\,U^{ce}(x')\,U^{db}(z)\,U^{ah}(w)$ | $+$ | $+2$ |
| $\mathsf{N}_7$ | III.III (7) | $+2$ | $f^{abc}\,\rho^e(x')\{\rho^h(x),\rho^d(y)\}\,U^{bd}(z)\,U^{ae}(x')\,U^{ch}(w)$ | $-$ | $-2$ |
| | | | | **sum for $\mathsf{N}_7$:** | $2-2=0$ |
| $\mathsf{N}_8 =$ | $f^{abc}\,[U(z)\rho(y)]^a\,[U(x)\rho(x)]^b\,[U(x')\rho(x')]^c$ | | | | |
| $\mathsf{N}_8$ | III.II (16) | $-2$ | $f^{abc}\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')$ | $-$ | $+2$ |
| $\mathsf{N}_8$ | III.VI (8) | $-2$ | $f^{abc}\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{ae}(x')\,U^{ch}(x)\,U^{bd}(z)$ | $+$ | $-2$ |
| | | | | **sum for $\mathsf{N}_8$:** | $2-2=0$ |

### 4.3 Four Wilson lines

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1 =$ | $f^{abc}\,\rho^a(x')\,[U(w')U(x)\rho(x)]^b\,[U(z)U(y)\rho(y)]^c$ | | | | |
| $\mathsf{N}_1$ | I.IV (13) | $+2$ | $f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)$ | $+$ | $+2$ |
| $\mathsf{N}_1$ | II.I (14) | $+2$ | $f^{abc}\,\rho^c(x')\{\rho^h(x),\rho^l(y)\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)$ | $+$ | $+2$ |

| net | source | $c$ | colour factor as written | $s$ | $c\,s$ |
|---|---|---|---|---|---|
| $\mathsf{N}_1$ | III.I (13) | $-2$ | $f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')$ | $+$ | $-2$ |
| $\mathsf{N}_1$ | III.II (13) | $-2$ | $f^{abc}\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ha}(w')$ | $+$ | $-2$ |
| | | | | **sum for $\mathsf{N}_1$:** | $2+2-2-2=0$ |
| $\mathsf{N}_2 =$ | $f^{abc}\,\rho^a(x')\,\big[U(w')U(w)\rho(x)\big]^b\,\big[U(z)U(y)\rho(y)\big]^c$ | | | | |
| $\mathsf{N}_2$ | I.IV (14) | $-2$ | $f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)$ | $+$ | $-2$ |
| $\mathsf{N}_2$ | III.I (15) | $+2$ | $f^{abc}\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')$ | $+$ | $+2$ |
| | | | | **sum for $\mathsf{N}_2$:** | $-2+2=0$ |

## 5 Numerical check

Each term is reduced to its symmetrised charge part. The three charges are placed on three points $p_1, p_2, p_3$ in all 3! ways, the terms are collected by kernel monomial, and the colour factors are evaluated with random SU(3) adjoint Wilson lines and random charges (three configurations). Script: `check_cancel.py`.

| Wilson lines | set | monomials | non-cancelling (3 configurations) | largest residual / largest term |
|---|---|---|---|---|
| $U = U^T$ (your convention) | all | 24 | 0, 0, 0 | $2.4e-15$ / $2.2e+00$ |
| $U = U^T$ (your convention) | $\alpha$ | 6 | 0, 0, 0 | $1.9e-15$ / $1.0e+00$ |
| $U = U^T$ (your convention) | $\alpha'$ | 6 | 0, 0, 0 | $1.9e-15$ / $1.1e+00$ |
| $U = U^T$ (your convention) | $\beta$ | 6 | 0, 0, 0 | $2.4e-15$ / $2.1e+00$ |
| $U = U^T$ (your convention) | $\beta'$ | 6 | 0, 0, 0 | $2.2e-15$ / $2.2e+00$ |
| general $U$, as written | all | 24 | 24, 24, 24 | $6.0e+00$ / $2.8e+00$ |
| general $U$, as written | $\alpha$ | 6 | 6, 6, 6 | $4.6e+00$ / $1.4e+00$ |
| general $U$, as written | $\alpha'$ | 6 | 6, 6, 6 | $4.5e+00$ / $1.1e+00$ |
| general $U$, as written | $\beta$ | 6 | 6, 6, 6 | $6.0e+00$ / $2.3e+00$ |
| general $U$, as written | $\beta'$ | 6 | 6, 6, 6 | $3.6e+00$ / $2.8e+00$ |

With a general $U$ and every chain read as $U(p_1)\cdots U(p_n)\rho$ (oriented from $f$ to the charge), the sum over the terms of each class vanishes to $10^{-14}$ (`general_u.py`).

**Reproducing.** `python3 parse3.py` (reads the list; checks that every index occurs twice and every kernel matches its label), `python3 check_cancel.py`, `python3 networks3.py`, `python3 general_u.py`, `python3 cmp_v2.py` (comparison with the corrected list of `three_rho_v2`), `python3 make_note3.py`; then `pdflatex ThreeRho_cancellation_v3.tex` twice.